

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Enabling Receiver on Asymmetric Transition
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Changing the conditions when activating the receiver in the Asymmetric <-> Symmetric transitions so it won't include detection of Gen 4 TS1 before TxFFE negotiation is executed
--

Benefits as a result of the changes:

Allowing an easier mechanism to activate the new receiver

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
--

None

An analysis of the hardware implications:
--

None

An analysis of the software implications:
--

None

An analysis of the compliance testing implications:
--

None

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Actual Change

(a). 4.2.2.5.2 – Transition from Symmetric Link to 1 Transmitter and 3 Receivers

To Text:

7. When the USB4 Port detects LFPS on Rx2, it shall do the following:
 - a. The USB4 Port shall transmit 24 CL0s_EXIT Ordered Sets with the *CL0s Phase* field set to 00t on its active transmitter at the beginning of an RS-FEC block.
 - b. ~~When the new receiver (Rx2) detects Gen 4 TS1, the USB4 Port~~ The new receiver (Rx2) shall enable high speed signal reception tCLxIdleRx time after the end of the last LFPS cycle is received and shall execute the Gen 4 receiver flow for TxFFE negotiation as described in Section 4.1.2.5.1.2.2. The receiver shall keep the *New Request* value of the Link Partner's transmitters (Tx0 and Tx1) set to 0b.

(b). 4.2.2.6.1 – Transition from 3 Transmitters and 1 Receiver to Symmetric Link

To Text:

7. When the USB4 Port detects LFPS on Rx1 it shall do the following:
 - a. The USB4 Port shall transmit 24 CL0s_EXIT Ordered Sets with the *CL0s Phase* field set to 00t. The 24 CL0s_EXIT Ordered Sets shall be transmitted on the Port's active transmitters at the beginning of an RS-FEC block.
 - b. ~~When it detects Gen 4 TS1, the USB4 Port~~ The new receiver (Rx1) shall enable high speed signal reception tCLxIdleRx time after the end of the last LFPS cycle is received and shall execute the Gen 4 receiver flow for TxFFE negotiation as described in Section 4.1.2.5.1.2.2. The receiver shall keep the *New Request* value of the existing transmitter (Tx0) set to 0b.